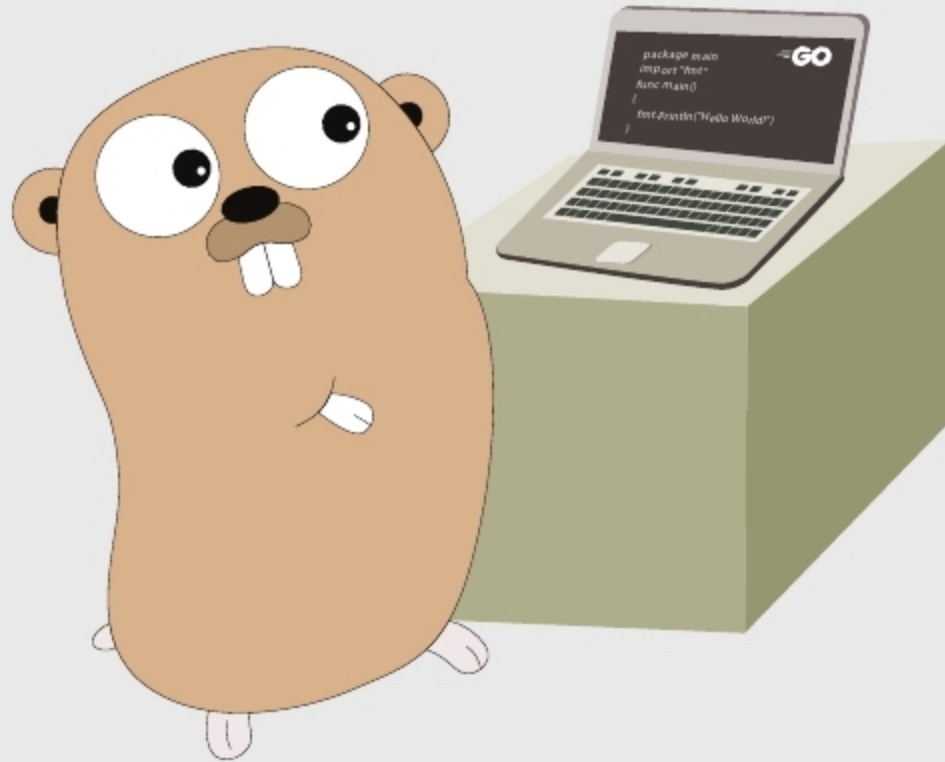


Golang: A Systems Programming Open Source Language by Google



Go is an attempt to combine the ease of programming of an interpreted, dynamically typed language with the efficiency and safety of a statically typed, compiled language. Go, often referred to as Golang, is an open source programming language created at Google in 2009 by Robert Griesemer, Rob Pike, and Ken Thompson.

Google describes its plans for Go in one of its blog posts as, “We’re hoping Go turns out to be a great language for systems programming with support for multi-processing and a fresh and lightweight take on object-oriented design, with some cool features like true closures and reflection. Go attempts to combine the development speed of working in a dynamic language like Python with the performance and safety of a compiled language like C or C++.”

This astounding language is meant to be simple-to-learn, concise and easy-to-read by other developers. It offers an expressive type system, as well as fast compilation, good performance and built-in language features that simplify concurrency. Go is reminiscent of C in its syntax, making it relatively easy for long-time C developers to learn. It uses interfaces as the building blocks of code reusability. Golang is a compiled language that supports static linking, and Go code can be statically linked to a native binary and be deployed in cloud servers easily, without worrying about dependencies.

Go, which provides a rich standard library, is designed primarily for systems programming. Its credentials are unsurpassed. Go provides type safety, garbage collection, dynamic-typing capability, many advanced built-in types such as variable length arrays and key-value maps, etc.

Why Go?

Go is recommended for the following features:

- It is simple, concurrent and has good cross-platform support.

- Go is awesome for microservices. Microservices designed with Golang are easy to modify, debug and test. You can change one service without impacting the entire functionality, and can quickly change the design and add new functions.
- Golang can be used for server development. It is a general-purpose language that is suitable for many things including server side Web applications, distributed apps, DevOps, etc. Currently, Docker is probably the most successful, well-known container platform. Google is also building Kubernetes, a Docker competitor using Go. Many big companies including Google, Dropbox, Soundcloud, Cloudflare, etc, use it for their projects.
- The use of Go for enterprise development has become a trend. For example, Netflix uses Go for exactly this purpose. Another reason is its efficient data processing on servers and optimised server loading. Golang enables non-disruptive background loading.

Go does have a few drawbacks, though, which are mainly in graphical and machine learning. It has become more prominent after many blockchain platforms started using it for development.

Installation

Golang is supported on all the three platforms — Mac, Windows and Linux. You can download the binary for the corresponding platform from <https://golang.org/dl/>.

In Linux, download the tar file from <https://golang.org/dl/> and unzip it to `/usr/local` to create a Go tree in `/usr/local/go`.